DEPARTMENT OF COMPUTER SCIENCE & APPLICATIONS
PANJAB UNIVERSITY, CHANDIGARH - 160014.
Website:http://dcsa.puchd.ac.in

Prof.R.K.Singla
Chairperson
Phone 2534075, 2534061
Email : chairpersondcsa@pu.ac.in

No._____/DCSA

Dated : _____/2019

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Notice

The Viva–Voce Examination in respect of Ms. Rohini Sharma,, on the subject matter of his Ph.D. thesis entitled**, "FLOW BASED ANOMALY DETECTION AND BEHAVIORAL ANALYSIS IN COMPUTER NETWORKS"** has been fixed **on March 22, 2019 at 12.00** noon in the PC Lab.-II of the Department of Computer Science and Applications, Panjab University, Chandigarh.

As per decision of the syndicate, vide Para 19 dated 31-08-1995, the Ph. D. thesis Viva–Voce examination in the various faculties is to be made public. Accordingly, we invite all interested persons to participate in the above mentioned Viva–Voce test/presentation **on March 22, 2019 at 12.00 noon in the PC Lab –II in the Department of Computer Science & Applications, P.U. Chandigarh**.

Chairperson

Copy for Information to:-

1. S.V.C ,PU., Chandigarh
2. D.U.I ,PU., Chandigarh
3. The C.O.E, PU., Chandigarh
4. The Director, APJ Abdul Kalam Computer Centre, PU
5. Director Research, PU., Chandigarh
6. D.R. (Secrecy)
7. Chairpersons, Maths., Physics, Zoology, Biochemistry, Biotechnology, Botany, Chemistry, UICET, Statistics, Geology and UIET, P.U. Chd.
8. The Supervisor -I, Prof. R.K. Singla, DCSA, PU
9. The Supervisor-II, Dr. Ajay Guleria, Deptt. of Computer Centre , PU (on leave)
10. Ms. Rohini Sharma, DCSA, the candidate
11. All the Faculty members /Research Scholars of DCSA
12. External examiner.
13. Notice Board of the DCSA.

DEPARTMENT OF COMPUTER SCIENCE & APPLICATIONS
PANJAB UNIVERSITY, CHANDIGARH - 160014.

Website:http://cs.puchd.ac.in

No._____/DCSA

Prof R.K.Singla
Chairperson.
Phone (O) 2534073

Dated : /2019

Professor Vikram Singh,
Department of Computer Science,
Choudhary Devi Lal University,
Sirsa, Haryana 125 055.

Dear Professor Vikram Singh,

As discussed with you, the Ph.D. viva voce examination of Ms. Rohini Sharma, on the subject matter of her Ph.D. thesis entitled, **, "FLOW BASED ANOMALY DETECTION AND BEHAVIORAL ANALYSIS IN COMPUTER NETWORKS"** has been fixed **on March 22, 2019 at 12.00 noon** in the PC Lab.-II of the Department of Computer Science and Applications, Panjab University, Chandigarh.

I shall be grateful if you kindly spare your valuable time to visit us and act as an Examiner for the same. Please inform us, if the arrangement of your stay in the P.U. Guest House is required or not.

Kindly communicate your programme to enable us to make arrangements.

Thanking you,

Yours sincerely,

(**R. K. Singla**)

DEPARTMENT OF COMPUTER SCIENCE AND APPLICATIONS
PANJAB UNIVERSITY, CHANDIGARH

Dated : 22.03.2019

Certified that the Ph.D. viva voce examination of Ms. Rohini Sharma was conducted on March 22, 2019 at 12.00 noon in the DCSA, P.U. Chandigarh in accordance with the Syndicate decision dated 31.8.1995.

The Ph.D. viva voce was made public.

(Prof. Vikram Singh)
External Examiner

(Prof. Indu Chhabra)
Expert

(Prof. R.K. Singla)
Supervisor –I

(Dr. Ajay Guleria)
Supervisor-II.

DEPARTMENT OF COMPUTER SCIENCE AND APPLICATIONS
PANJAB UNIVERSITY, CHANDIGARH

Dated: 22.03.2019

Thesis Presentation Certificate

Certified that Ms. Rohini Sharma, has made a successful presentation of his Ph.D. thesis entitled, **, "FLOW BASED ANOMALY DETECTION AND BEHAVIORAL ANALYSIS IN COMPUTER NETWORKS"** on March 22, 2019 at 12.00 noon. in the PC Lab.- II of the Department of Computer Science and Applications, Panjab University, Chandigarh.

(Prof. Vikram Singh)
External Examiner

(Prof. Indu Chhabra)
Expert

(Prof. R.K. Singla)
Supervisor –I

(Dr. Ajay Guleria)
Supervisor-II.

DEPARTMENT OF COMPUTER SCIENCE AND APPLICATIONS
PANJAB UNIVERSITY, CHANDIGARH

Dated: 22.03.2019

The Assistant Registrar (Secrecy)
Panjab University
Chandigarh.

Subject : Report of viva-voce examination of Ms. Rohini Sharma.

Dear Sir,

This has reference to your Office letter No. 7412/Secrecy/Theses dated 28.02.2019 on the subject cited above.

Ms. Rohini Sharma has made a successful presentation of his Ph.D. thesis entitled, **"FLOW BASED ANOMALY DETECTION AND BEHAVIORAL ANALYSIS IN COMPUTER NETWORKS"** on March 22, 2019 at 12.00 noon in the PC-Laboratory-II (Top Floor) of the Department of Computer Science and Applications, Panjab University, Chandigarh by the following Board of Examiners:

1. Professor Vikram Singh,
   Department of Computer Science,
   Choudhary Devi Lal University,
   Sirsa, Haryana 125 055

2. Prof. R. K. Singla, (Supervisor -I)
   Department of Computer Science and Applications,
   P.U. Chandigarh

3. Dr. Ajay Guleria, (Supervisor -II)
   Department of Computer Science and Applications,
   Panjab University, Chandigarh.

4. Professor Indu Chhabra, (Expert)
   Department of Computer Science and Applications,
   Panjab University, Chandigarh.

Please find enclosed the following documents along with the reports of the thesis examiners:

1. Report of the Board of Examiners of viva-voce.
2. Correction Certificate
3. Thesis Presentation Certificate
4. Certificate that Ph.D. viva voce was made public as per decision of the Syndicate meeting dated 31.8.1995.
5. Ph.D. Examiners' Report (in original)

Yours faithfully,

(R. K. Singla)
Chairperson

DEPARTMENT OF COMPUTER SCIENCE AND APPLICATIONS
PANJAB UNIVERSITY, CHANDIGARH

Dated: 22.03.2019

CORRECTION CERTIFICATE

Certified that no change/correction was suggested to be made by the examiners in the Ph.D. thesis entitled, **"FLOW BASED ANOMALY DETECTION AND BEHAVIORAL ANALYSIS IN COMPUTER NETWORKS" of Ms. Rohini Sharma**

(Prof. Vikram Singh)
External Examiner

(Prof. Indu Chhabra)
Expert

(Prof. R.K. Singla)
Supervisor –I

(Dr. Ajay Guleria)
Supervisor-II.

DEPARTMENT OF COMPUTER SCIENCE AND APPLICATIONS

PANJAB UNIVERSITY, CHANDIGARH

Ph.D. Viva Voce Examination
Report of the Board of Examiners

The public viva-voce examination of Ms. Rohini Sharma of her Ph.D. thesis entitled, **"FLOW BASED ANOMALY DETECTION AND BEHAVIORAL ANALYSIS IN COMPUTER NETWORKS"** in the subject of Computer Science was conducted on March 22, 2019 at 12.00 noon in PC-Laboratory - II (Top Floor) of the Department of Computer Science and Applications, Panjab University, Chandigarh.

The following members of the Board were present:

1. Professor Vikram Singh,
   Department of Computer Science,
   Choudhary Devi Lal University,
   Sirsa, Haryana 125 055

2. Prof. R. K. Singla, (Supervisor -I)
   Department of Computer Science and Applications,
   P.U. Chandigarh

3. Dr. Ajay Guleria, (Supervisor -II)
   Department of Computer Science and Applications,
   Panjab University, Chandigarh.

4. Professor Indu Chhabra, (Expert)
   Department of Computer Science and Applications,
   Panjab University, Chandigarh.

Report:

(Vikram Singh)	(Indu Chhabra)	(R.K. Singla)	(Ajay Guleria)
External Examiner	Expert	Supervisor –I	Supervisor –II

DEPARTMENT OF COMPUTER SCIENCE AND APPLICATIONS

PANJAB UNIVERSITY, CHANDIGARH

Dated: 22.03.2019

To Whom It May Concern

Ms. Rohini Sharma, appeared for the viva (Public) examination for the award of Ph.D. Degree on March 22, 2019 at 12.00 noon in the Department of Computer Science and Applications, Panjab University, Chandigarh and her performance was satisfactory. The title of her Ph.D. thesis is **"FLOW BASED ANOMALY DETECTION AND BEHAVIORAL ANALYSIS IN COMPUTER NETWORKS"**. Her name is under consideration for the award of Ph.D. Degree.

(R. K. Singla)
Chairperson

DEPARTMENT OF COMPUTER SCIENCE & APPLICATIONS
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Phone 2534075, 2534061
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No._____/DCSA

Dated : _____/2019

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Professor Vikram Singh, Department of Computer Science, Choudhary Devi Lal University, Sirsa, Haryana will be visiting the Department of Computer Science and Applications, P.U., Chandigarh on March 22, 2019 as an Examiner to conduct the Ph.D. Viva Voce Examination in respect of Ms. Rohini Sharma scheduled to be held on 22.03.2019 at 12.00 noon on the subject matter of her Ph.D. thesis entitled **"FLOW BASED ANOMALY DETECTION AND BEHAVIORAL ANALYSIS IN COMPUTER NETWORKS"** in the faculty of Science, by his own Car or Taxi from Delhi to Chandigarh on March 22, 2019.

The Controller of Examinations is requested to kindly accord permission to Professor Vikram Singh to travel by own Car or Taxi on March 22, 2019 and pay him TA/DA + actual toll charges (UP and down) from Sirsa to Chandigarh and back, who is visiting the DCSA to conduct the Ph.D. Viva Voce Examination as per the orders of the Vice-Chancellor conveyed vide letter No. 7412/Secy/Thesis dated 28.02.109 (copy attached).

(R. K. Singla)

Controller of Examinations